

## Unit 12, Lesson 3: Interpreting Data Distributions for Numerical Data



### Benchmarks

MA.912.DP.1.1 Given a set of data, select an appropriate method to represent the data, depending on whether it is numerical or categorical data and on whether it is univariate or bivariate.

MA.912.DP.1.2 Interpret data distributions represented in various ways. State whether the data is numerical or categorical, whether it is univariate or bivariate and interpret the different components and quantities in the display.



### Learning Targets

- ☐ I can interpret data distributions from an appropriate display for univariate, numerical data.
- ☐ I can interpret data distributions from an appropriate display for bivariate, numerical data.

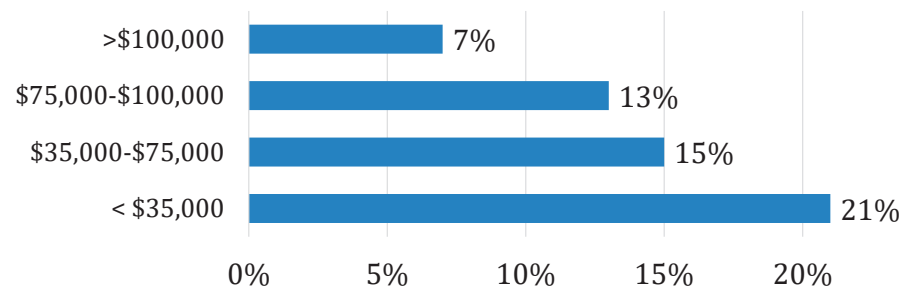


### 12.3.1 Warm-Up: Smoking and Income

In 2019, about 14% of U.S. adults smoked cigarettes, a decline from about 20% in 2005.

The bar graph shows the percent of smokers by household income.

**Smokers by Household Income**



1. What do you notice?

2. What do you wonder?



### 12.3.2 Refresher: Measures of Center and Variation



**Measures of center**, such as **mean** and **median**, are used to describe a typical value in the data set.



**Measures of variation**, such as **range** and **interquartile range**, are used to describe how “spread out” the data is.



The **mean** is the arithmetic average of a set of numbers. It is a measure of center, or central tendency.



The **median** is the middle of an ordered list of the values. If the list has an odd number of values, it is the middle value of that list. If the list has an even number of values, it is the average of the two middle values. It is a measure of center, or central tendency.



The **range** is the difference between the highest data value and the lowest data value in a data set. It is a measure of variation.



A measure of variation in a set of numerical data, the **interquartile range (IQR)** is the distance between the first and third quartiles of the data set.

1. Use the word bank to complete the following. Not all terms will be used.

mean      median      mode      range  
interquartile range      quartile



When a distribution is bell-shaped, the \_\_\_\_\_ is the preferred measure of center and the \_\_\_\_\_ is the preferred measure of variation to describe the distribution. However, the context must be considered.

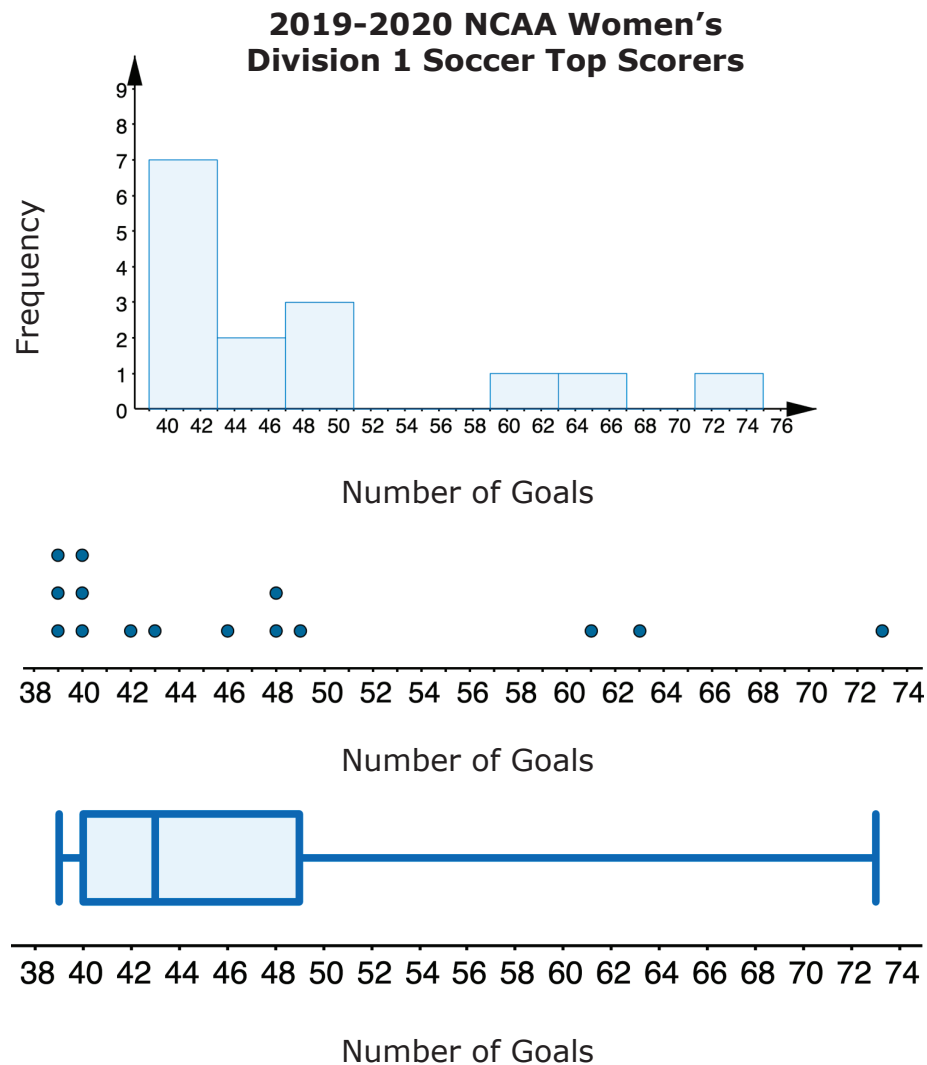


When outliers are present or a distribution is skewed, the \_\_\_\_\_ is the preferred measure of center and the \_\_\_\_\_ is the preferred measure of variation to describe the distribution. However, the context must be considered because outliers may be important in understanding the data.



### 12.3.3 Guided Practice: Comparing Distributions

1. Shown are a histogram, line plot, and box plot of the number of goals scored by the top 15 women goal scorers during the 2019-2020 NCAA Women's Division 1 soccer season.



- a. The ☐ histogram ☐ line plot ☐ box plot can be used to determine the mean.
- b. The ☐ histogram ☐ line plot ☐ box plot can be used to determine the median.
- c. What is the appropriate measure of center for the data set?
- d. Which display would you prefer to use to determine the median?
- e. Interpret the median of the data set.

f. The ☐ histogram ☐ line plot ☐ box plot can be used to determine the range.

g. The ☐ histogram ☐ line plot ☐ box plot can be used to determine the interquartile range.

h. What is the appropriate measure of spread for the data set?

i. Which display would you prefer to use to determine the interquartile range?

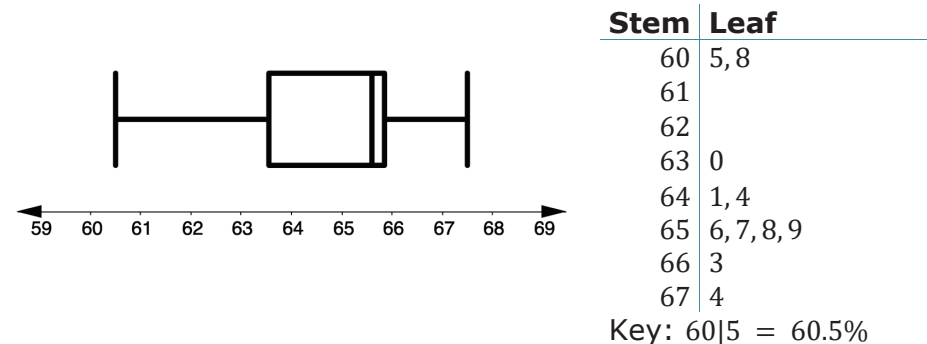
j. Interpret the interquartile range.



### 12.3.4 Collaboration: Univariate vs. Bivariate Data

- Tom Brady is considered by most to be one of the best NFL quarterbacks of all time. His pass-completion percentages from 2009 to 2019 are shown in the box plot and stem-and-leaf plot.

#### Tom Brady's Completion Percentages 2009-2019

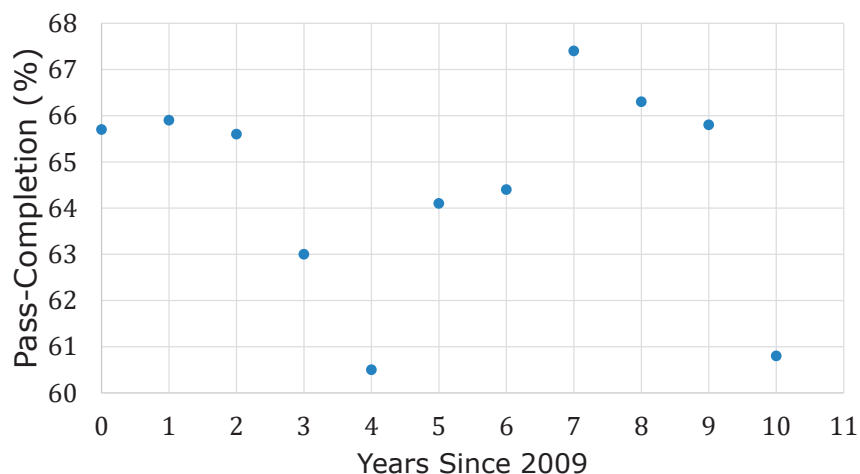


- The ☐ mean ☐ median best summarizes the center of the data. The ☐ mean ☐ median is not the best measure of center because it is influenced by the cluster of values, ☐ 60.5% and 60.8%, ☐ 66.3% and 67.4%, that are far from the rest of the data values.
- The ☐ IQR ☐ range best summarizes the variation of the data.

2. Tom Brady's pass-completion percentages from 2009 to 2019 are listed in the table and shown in the scatter plot.

| Year | 2009 | 2010 | 2011 | 2012 | 2013 | 2014 |
|------|------|------|------|------|------|------|
| %    | 65.7 | 65.9 | 65.6 | 63   | 60.5 | 64.1 |
| Year | 2015 | 2016 | 2017 | 2018 | 2019 |      |
| %    | 64.4 | 67.4 | 66.3 | 65.8 | 60.8 |      |

**Tom Brady's Pass-Completion 2009-2019**



- a. The scatter plot displays the relationship between

two ☐ numerical ☐ categorical variables. Another display that

could be created for this ☐ univariate ☐ bivariate data set

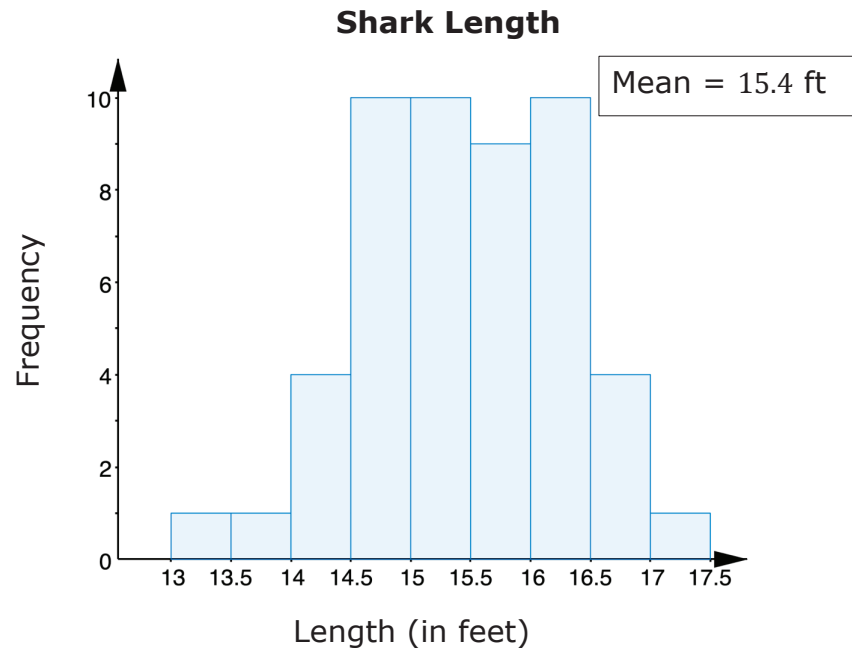
is a ☐ line graph. ☐ histogram.

- b. What is one observation you can make from the scatter plot that you could not from the displays summarizing only the pass-completion percentages?



### 12.3.5 Guided Instruction: Summarizing Data

1. The histogram summarizes the lengths, in feet (ft), of a group of adult female sharks.



- a. Describe the shape of the distribution.
- b. Interpret the mean of the distribution by completing the following statement.

The typical length of an adult female shark is \_\_\_\_\_.

- c. Interpret the variation in shark length.

Another measure of variation that summarizes bell-shaped numerical data is **standard deviation**.

The **standard deviation** of a data set is the typical variation of each value to the mean. The larger the standard deviation, the more spread out the data is.

- d. The standard deviation of shark length is 0.75 ft. Interpret the standard deviation.
- For this sample of adult female sharks, the typical deviation from the mean of \_\_\_\_\_ is \_\_\_\_\_. This group of sharks have lengths that are relatively similar.



When summarizing data, both a measure of center and a measure of variability should be used to provide a summary of both what is typical of the data (center) *and* how far the data is spread from that value (variability).

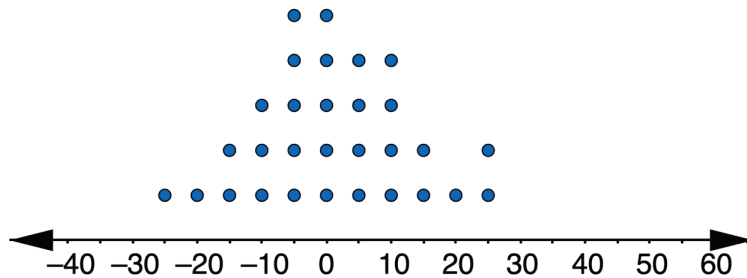
- When using the **mean** to summarize the center, **standard deviation** should be used to describe the spread when the data is bell-shaped.
- When using the **median** to summarize the center, **interquartile range** should be used to describe the spread.



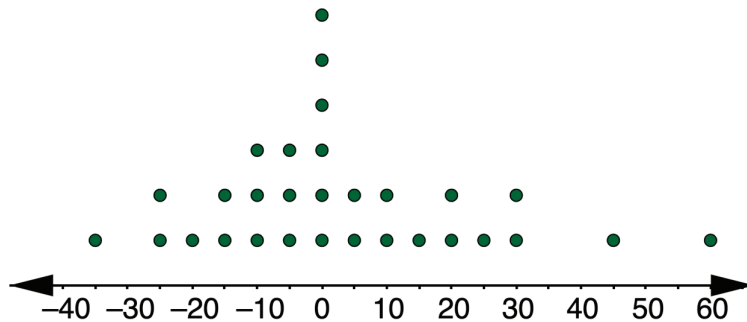
### 12.3.6 Try It: Summarizing Data

1. The dot plots display the arrival times of 30 flights for two different airlines.

**Arrival Times of Airline P**



**Arrival Times of Airline Q**



A negative number represents the number of minutes the flight arrived before its scheduled time.

A positive number represents the number of minutes the flight arrived after its scheduled time.

A zero indicates that the flight arrived at its scheduled time.

- a. Which airline do you expect to have a larger standard deviation?

- b. Based on these data, from which airline would you choose to buy your ticket? Use measure(s) of center and variation to justify your choice.



### 12.3.7 Wrap-Up: Reflection

1. What was the **muddiest** (most difficult or confusing) point in today's lesson?



## Unit 12, Lesson 4: Displaying and Interpreting Univariate Categorical Data



### Benchmarks

MA.912.DP.1.1 Given a set of data, select an appropriate method to represent the data, depending on whether it is numerical or categorical data and on whether it is univariate or bivariate.

MA.912.DP.1.2 Interpret data distributions represented in various ways. State whether the data is numerical or categorical, whether it is univariate or bivariate and interpret the different components and quantities in the display.



### Learning Targets

- ☐ I can create an appropriate graphical display for univariate categorical data.
- ☐ I can interpret a graphical display for univariate categorical data.